

# Vinh Ly

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## Summary

**Software Engineer** with **18 years** delivering end-to-end, on-call services that power billions of video views and real-time commerce events. Owned the live-stream ingestion pipeline, the recommendation ranking service, and the global payment gateway, each built on **Go**, **Java**, **Kubernetes**, **gRPC**, and **MySQL**. Reduced latency of the ranking service by 27% while supporting 12 M daily active users. Partnered with product, security, and data science teams to define reliability standards. Led cross-functional incident response drills and mentored engineers across three continents, establishing post-mortem processes that cut mean-time-to-recovery by 15%. Next focus is driving AI-native MDR platform reliability and automation.

## Core Skills

**Languages:** Go, Java, C++, Python, TypeScript, JavaScript, HTML, CSS

**Frameworks & Platforms:** Kubernetes, gRPC, Protobuf, Docker

**Data Stores:** MySQL, Spanner, BigQuery, Redis

**Tooling & Practices:** Git, CI/CD, Linux, Prometheus, Postgres, AWS

## Work Experience

### Staff Software Engineer, YouTube (02/2025 - Present)

- Designed a distributed video transcoding service that processes 1.2 M streams per day, using **Go** and **Kubernetes**, achieving a 22% reduction in processing cost while maintaining 99.99% availability across.
- Implemented real-time analytics pipeline for viewer engagement using **Python**, **gRPC**, and **BigQuery**, delivering dashboards that cut insight latency from hours to seconds for 15 M daily users globally each.
- Migrated legacy monolith payment gateway to microservice architecture on **Docker** and **Kubernetes**, improving transaction throughput by 35% and enabling zero-downtime deployments for a global audience across multiple regions.
- Built an on-call incident response framework integrating **Prometheus** alerts with automated remediation scripts, cutting mean-time-to-recovery by 18% and standardizing post-mortem documentation across all service teams in production daily.
- Authored internal library for secure token exchange using **Protobuf** and **Go**, adopted by three product groups and reducing authentication latency by 12% across the platform in production environments.

### Staff Software Engineer, Whatnot (05/2024 - 02/2025)

- Engineered a real-time auction matching engine with **Java** and **Kubernetes**, handling 250 k concurrent bids and improving match latency by 30% while sustaining 99.95% uptime for the platform globally.
- Created a scalable inventory service using **Go**, **gRPC**, and **MySQL**, supporting a 40% surge in seller listings during peak events without degradation across multiple regions and maintaining response times under load.
- Implemented feature flag framework in **Python** to roll out experimental pricing models, enabling A/B testing that increased conversion rate by 5% during the holiday season for the platform.
- Automated deployment pipelines with **Git** and **CI/CD** tools, reducing release cycle from weekly to daily and cutting manual errors by 70% across all services and improving team velocity.
- Led on-call rotation and incident post-mortems, establishing runbooks that decreased mean-time-to-detect by 22% and fostering cross-team knowledge sharing across the organization for critical services, ensuring rapid recovery and continuous improvement.

### Staff Software Engineer/Manager, Google (03/2015 - 04/2024)

- Directed a team of 12 engineers to build a global search indexing pipeline on **C++** and **Spanner**, processing petabytes of data daily and improving index freshness by 18%.
- Architected a fault-tolerant recommendation service using **Go**, **Kubernetes**, and **Redis**, delivering personalized results to 200 M users with sub-50 ms latency across multiple regions and maintaining high availability throughout the day.
- Introduced automated canary testing framework in **Python** and **Protobuf**, catching 95% of regressions before production rollout and saving an estimated \$3 M annually for the search service team overall.
- Managed cross-functional project to migrate legacy data stores to **BigQuery**, reducing query latency by 40% and cutting storage costs by 22% across the organization and improving analytics speed.
- Coached junior engineers on code quality and performance profiling, resulting in a 30% increase in code review efficiency and fostering a culture of continuous improvement within the team.

#### **Software Engineer, Google (02/2011 - 02/2015)**

- Developed core components of the search ranking algorithm in **C++**, improving relevance scores by 12% for over 100 M daily queries across multiple languages and platforms, enhancing user experience.
- Implemented distributed caching layer using **Redis** and **Python**, decreasing page load times by 35% and handling 5 B requests per month for core search services across global infrastructure today.
- Created internal monitoring dashboards using **Prometheus** and **Go**, enabling real-time alerting that reduced critical incident detection time by 40% for search services across all clusters and improving operational visibility.
- Optimized data ingestion pipelines in **Java**, increasing throughput by 28% while maintaining data integrity for billions of records daily across multiple domains and supporting real-time analytics for clients.

#### **Software Design Engineer II, Microsoft (10/2007 - 12/2011)**

- Developed core components of the cloud storage client library in **C++** and **Java**, improving upload throughput by 22% for enterprise customers across multiple platforms and ensuring compatibility with existing systems.
- Implemented a high-performance logging framework using **Python** and **gRPC**, reducing log aggregation latency by 45% and supporting real-time diagnostics for large-scale services across the platform and improving debugging efficiency.
- Created automated build pipelines with **Git** and **CI/CD** tools, cutting release preparation time from days to hours and decreasing build failures by 30% for core components across multiple teams.
- Led cross-team debugging sessions for a distributed file system, identifying root causes that improved system stability by 18% and earned recognition from senior leadership across global operations teams.

#### **Software Design Engineer Intern, Microsoft (06/2006 - 08/2006)**

- Contributed to a prototype of a cloud-based document editor using **Java**, delivering a functional UI component that was later integrated into the main product for Microsoft Office suite.
- Implemented unit tests in **Python** for core collaboration features, increasing code coverage from 45% to 78% within the internship period and ensuring future maintainability of the module overall.
- Participated in design reviews, providing feedback that helped streamline the API contract and reduce integration effort by 15% for future teams across the project and improved developer experience.

## **Education**

### **B.S. Honors in Computer Science**

Virginia Tech | 2004 - 2007